pmsimstats-ng Documentation Index

pmsimstats team

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This document indexes the technical documentation for the `pmsimstats-ng` simulation framework. Documents are grouped by their role in two narratives: (a) the Hendrickson et al. (2020) research program on biomarker-moderated treatment response in aggregated N-of-1 trial designs, and (b) the development path from the original codebase to the current dual-architecture framework.

1 Research narrative

These documents describe the scientific goals, statistical methods, and simulation results for the Hendrickson et al. (2020) power analysis of predictive biomarker-treatment interactions.

Document	Scope
04	Revised power analysis with corrected covariance structure and carryover model (Architecture B). Includes commit-by-commit comparison appendix.
06	AR(1) residual correlation: transition from `lmer` to `nlme` with `corCAR1`. Applies to both architectures.
07	Positive definiteness failures in the MVN covariance matrix (Architecture B specific).
08	Derivation of the λ_{cor} BM-BR correlation decay during off-drug periods (Architecture B specific).
09	Analysis of the step-function BM-BR correlation artifact in the original code (Architecture B specific).
18	Response parameter sensitivity (Figure 5): original vs. revised implementations (Architecture B).

2 Dual-architecture framework

These documents describe the two data-generating process architectures for the biomarker-treatment interaction and the implementation of Architecture A (direct mean moderation).

Document	Scope
02	Two DGP architectures for biomarker-treatment interactions: theoretical comparison, empirical results, biological assumptions, and analysis strategies.
19	Architecture A implementation: four attempts, correct calibration, PD constraint removal, and Hendrickson publication parameter evaluation.

3 Codebase reference

Document	Scope
01	Codebase overview: repository structure, R package functions, analysis scripts, key parameters.
13	Figure 4 walkthrough: original publication methodology (*pre-audit historical reference*).

4 Collaborator deliverables

Document	Scope
03	Audit and revision report: status summary for collaborator presentation.
10	Carryover analysis model assessment: standard methodology and application to pmsimstats.
14	Trial analysis implementation guide: response to collaborator goals and priorities.

5 Internal development

Document	Scope
15	Tidyverse alignment plan: 9-step roadmap for matching the 2025 pipeline to the orig gold standard.
16	Tidyverse code review: dead code, duplication, and simplification opportunities.
17	Simplification plan: options for reducing complexity in the simulation framework.

6 Executable reports

The Gompertz clinician guide (`analysis/report/11-gompertz-clinician-guide.Rmd`) contains executable R code and is rendered via `rmarkdown`. It resides in the zzc report directory rather than in `docs/`.

7 Removed documents

Document	Reason
`05`	Duplicate of `04` (removed).
`11`	Moved to `analysis/report/` (executable Rmd).
`12`	Number not assigned.